Alexander Puck Neuwirth

Theoretical and Computational Particle Physicist

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github.com/APN-Pucky • gitlab.com/APN-Pucky • scholar.google.com/citations?user=qdaEI4YAAAAJ
INSPIRE-HEP • Codeberg • Milano-Bicocca faculty page

Education

University of Münster *Münster, Germany*
PhD in Theoretical Physics *2024*
- Thesis: "Precision Computations for Photons, Jets and BSM-Particles at the LHC"
- Advisors: Prof. Dr. Michael Klasen, Prof. Dr. Christian Klein-Bösing

Experience

Research Fellow, Department of Physics "Giuseppe Occhialini" *Milan, Italy*
University of Milano-Bicocca *2024 – present*
- Precision predictions for photon production, electroweak corrections and Monte Carlo event generators

Contract lecturer, Informatica 1 (academic year 2024/2025) *Milan, Italy*
University of Milano-Bicocca *2024 – 2025*
- Taught in English; course materials prepared in Italian

Publications

Photon Fragmentation Functions in LHAPDF: Existing Sets and a Pythia-Based Extraction Aug 2026
Alexander Puck Neuwirth
arxiv.org/abs/2608.15944 (arXiv preprint)

pylhe: A Lightweight Python Interface to Les Houches Event Files July 2026
Alexander Puck Neuwirth, Matthew Feickert, Lukas Heinrich, Eduardo Rodrigues
10.21105/joss.11128 (Journal of Open Source Software, 11(126), 11128)

Conversion of Photons to Dileptons in the Kroll-Wada and Parton Shower Approaches June 2025
Tomáš Ježo, Michael Klasen, *Alexander Puck Neuwirth*
10.1007/JHEP03(2026)024 (JHEP 03 (2026) 024)

Prompt Photon Production with Two Jets in POWHEG Sept 2024
Tomáš Ježo, Michael Klasen, *Alexander Puck Neuwirth*
10.1007/JHEP02(2025)125 (JHEP 02 (2025) 125)

Combination and Reinterpretation of LHC SUSY Searches Mar 2024
Alexander Feike, Juri Fiaschi, Benjamin Fuks, Michael Klasen, *Alexander Puck Neuwirth*
10.1007/JHEP07(2024)122 (JHEP 07 (2024) 122)

Di-Electron Production at the LHC: Unravelling Virtual-Photon and Heavy-Flavour Contributions Jan 2024
Anton Andronic, Tomáš Ježo, Michael Klasen, Christian Klein-Bösing, *Alexander Puck Neuwirth*
10.1007/JHEP05(2024)222 (JHEP 05 (2024) 222)

Electroweak Superpartner Production at 13.6 TeV with Resummino Apr 2023
J. Fiaschi, B. Fuks, M. Klasen, *A. Neuwirth*
10.1140/epjc/s10052-023-11888-y (Eur. Phys. J. C 83, 707)

Soft Gluon Resummation for Associated Squark-Electroweakino Production at the LHC Feb 2022
J. Fiaschi, B. Fuks, M. Klasen, *A. Neuwirth*
10.1007/JHEP06(2022)130 (JHEP 06 (2022) 130)

Associated Squark-Electroweakino Production with NLO+NLL Precision May 2022
Juri Fiaschi, Benjamin Fuks, Michael Klasen, *Alexander Neuwirth*
arxiv.org/abs/2205.07298 (Proceedings, 56th Rencontres de Moriond (QCD session))

Open Source

- **pylhe** — core maintainer; lightweight Python interface to Les Houches Event files (Scikit-HEP)
- **LHEUTILS** — author; command-line utilities for LHE event processing
- **LHAPDF** — code contributor to the standard LHC parton-distribution-function library
- **HepMC, Rivet, YODA** — contributor to the shared Monte Carlo event-record and analysis toolchain
- **Gentoo Linux** — developer, maintaining scientific packages in the gentoo-sci overlay

Skills

Programming: C/C++, Fortran, Python, Rust, CUDA, Bash

HEP software: POWHEG-BOX(-RES), PYTHIA 8, GENEVA, Rivet, YODA, FastJet, HepMC, LHAPDF, OpenLoops, RECOLA, MadGraph

Scientific computing: Monte Carlo integration and event generation, adaptive/rejection sampling, HPC and Slurm cluster workflows, reproducible packaging and CI